Candidate Name _____

Centre Number	Candidate Number

EXAMINATIONS COUNCIL OF ZAMBIA

Examination for School Certificate Ordinary Level

Design and Technology 6045/1

Paper 1 (Theory & Design)

Specimen

Additional Materials
A3 Drawing Paper
Drawing instruments

Time: 2 hours 30 minutes **Marks: 110**

Instructions to Candidates

Write your **name**, **centre number** and **candidate number** in the spaces at the top of this page and on the separate **Answer Booklet provided**.
If working is needed for any question, it must be shown in the space provided in the question paper and answer Booklet.
There are **two** sections in this paper.

Section A
Answer **all** questions.
Write your answers in the spaces provided on the question paper.

Section B
Part I is compulsory
Part II has **three** questions, answer **one** question.
Part III: has two questions, answer **one** question.
Write your answers in the Answer Booklet provided.
Where a freehand sketch is required, it must be drawn to half full size with good linework.
At the end of the examination:
1 fasten the Answer Booklet used securely to the question paper,
2 enter the numbers of the Section B questions you have answered in the grid under column, "For **candidate use**".

For Candidate use	For Examiner's use
Section A	
Section B	
Part I	
Part II	
Part III	
Total	

Information for candidates
The number of marks is given in brackets [ ] at the end of each question or part of question.
Use of colour is allowed.
All dimensions are in millimetres.
Cell phones and calculators are not allowed in the examination room.

©ECZ/SC/2020/K1
This question paper consists of 10 printed pages

ZEDPASTPAPERS.COM
Specimen
zedpastpapers.com
goto zedpastpapers.com for more

Section A [30 marks]

Answer all the question in the spaces provided on the question paper.
If working is needed for any question, it must be shown in the space below tha question.

1 **(a)** **Figure 1** shows one of the driving tools used in the Design and Technology workshop.

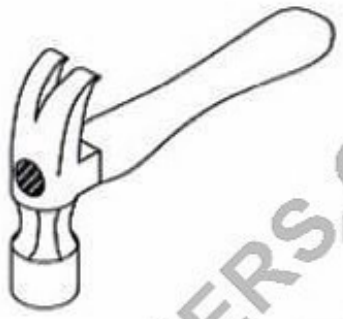


Figure 1

(i) Give the name of the tool in figure 1.

..... [1]

(ii) State **two** safety rules to be observed when using the tool in figure 1.

1 .....

.....

2 .....

..... [2]

(iii) Give **two** uses of the tool in figure 1 above.

1 .....

2 ..... [2]

zedpastpapers.com

(b) State **one** property of ash which makes it suitable for the mallet handle. [1]

[Total: 6 marks]

ZEDPASTPAPERS.COM
Specimen

goto zedpastpapers.com for more

2 **Figure 2** shows some components used in Design and Technology.

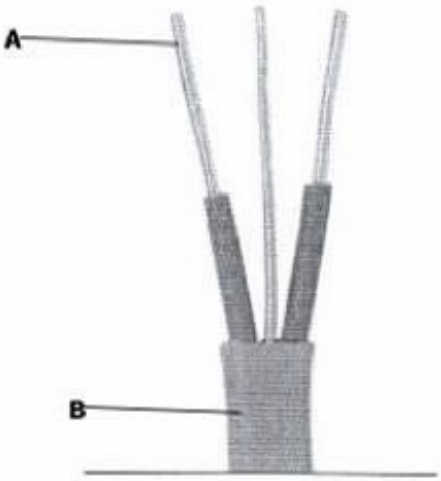


Figure 2

(a) **(i)** Identify the components for the parts labelled **A** and **B**.

A .....

B ..... [2]

(b) Pulleys and gears are both used in the transmission of motion.

(i) Give **one** advantage of gears over pulleys

.....

..... [2]

(ii) Mention **one** example where gears are used.

.....

..... [2]

[Total: 6 marks]

zedpastpapers.com

ZEDPASTPAPERS.COM
Specimen

goto zedpastpapers.com for more

3 **Figure 3** shows a cabinet to be used in the workshop for storing tools.



Figure 3

(a) Briefly describe the type of finish you would use for the cabinet.

.....

..... [1]

(b) Explain why safety rules should be observed when carrying out the following processes.

(i) Using a circular saw

..... [1]

(ii) Melting plastics

..... [1]

(c) **(i)** Define an entrepreneur

.....

..... [2]

zedpastpapers.com

(ii) What does the abbreviation SWOT stand for in marketing?

..... [1]

[Total: 6 marks]

ZEDPASTPAPERS.COM
Specimen

goto zedpastpapers.com for more

4 **Figure 4** shows a simple electric circuit. Study the drawing and answer the questions that follow.

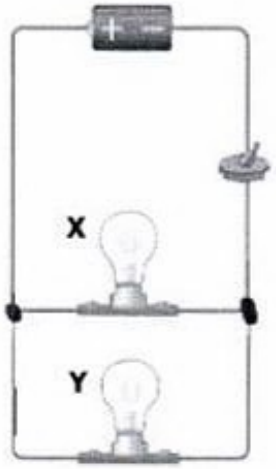


Figure 4

(a) How are the bulbs arranged in the diagram?

..... [1]

(b) When bulb **X** blows out, what happens to bulb **Y**?

..... [1]

(c) Using electronic symbols, redraw the circuit diagram in **figure 4**.

[4]

[Total: 6 marks]

5 **(a)** Explain the term static force.

zedpastpapers.com

.....

..... [2]

(b) Give **two** examples of forces.

1 .....

2 ..... [2]

ZEDPASTPAPERS.COM
Specimen
goto zedpastpapers.com for more

(c) State any **two** reasons that lead to structural failure

1 .....

..... [1]

2 .....

..... [1]

[Total: 6 marks]

Section B [80 marks]

There are three parts in this section. Part I, is compulsory. Part II and III, answer one question from each part.

Part I Compulsory

6 Study **Figure 5** and refer to it as you answer the questions that follow.



Figure 5

(a) **(i)** Write down the situation. [2]

(ii) Identify the problem from the situation in **figure 5**. [2]

(b) **(i)** Derive a design brief from the problem in **(a) (ii)** above. [2]

(ii) Using annotations, sketch one possible solution in pictorial view for the situation above. [12]

(c) List **three** specifications which could be considered when coming up with your design solution. [3]

(d) You are also expected to do research in order to gather information that will guide you in making decisions. Mention **four** possible areas of research. [4]

[Total 25 marks]

ZEDPASTPAPERS.COM

goto zedpastpapers.com for more

Part II Answer one question

7 **(a)** Jane specified the following materials for her Grade 12 artefact: aluminium, mukwa, acrylic

For each type of material;

(i) State **two** properties [6]

(ii) Give an example where each material can be used [3]

(iii) Identify the tools that can be used when cutting the materials stated [3]

(iv) The finish that can be applied to each of the materials [3]

(b) Most products we use have been fabricated (joining materials together) With the aid of annotated sketches, illustrate the stages of how to come up with any joint of your choice. [10]

[Total: 25 marks]

8 **(a)** **Figure 6** shows some fittings and fixings that are used in artefact making.



1 **2** **3** **4**

Figure 6

(i) Name the fittings [4]

(ii) Give an example where each of them can be used. [4]

(iii) Mention a tool that can be used when preparing to fix or fit each of them. [4]

(b) The drawing in **figure 7** shows a wooden box made by a Grade 7 learner which he wants to use for keeping coins.

zedpastpapers.com

goto zedpastpapers.com for more

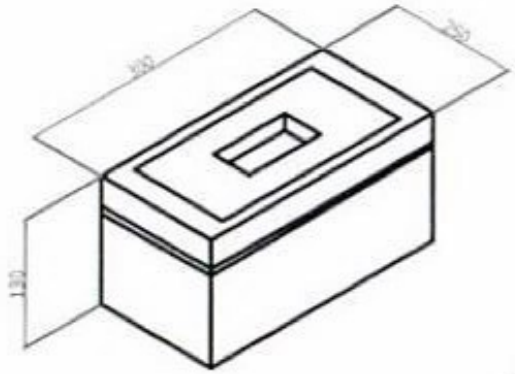


Figure 7

(i) Which of the four fixings in **(a)** above would the learner use on the artefact? [2]

(ii) With the help of sketches, suggest where the fixings you have mentioned in **(b)(i)** will be applied. [8]

(iii) Explain how the child will be removing the money. [3]

[Total: 25 marks]

9 **Figure 8** shows a work piece being drilled. The part marked **'X'** is slot for a key hole.

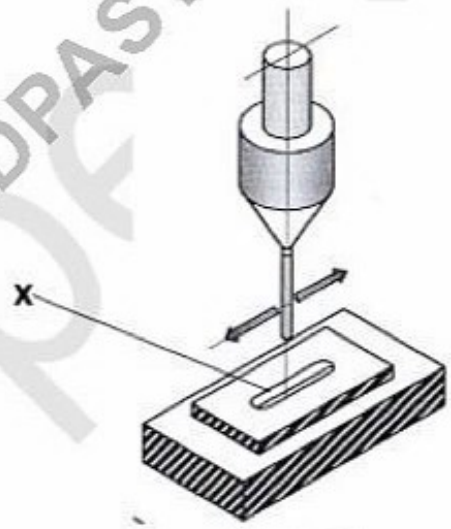


Figure 8

zedpastpapers.com

(a) List **two** reasons why it is necessary to place scrap wood underneath the work piece when drilling. [2]

(b) **(i)** Illustrate, with the aid of sketches, how you would obtain the slot **'X'**.[12]

(ii) Give examples of **two** materials you can make a slot in. [2]

(c) **(i)** Classify **three** sources of energy. [3]

goto zedpastpapers.com for more

(ii) Using sketches demonstrate the following motions:

1 Reciprocating [2]

2 Oscillating [2]

3 Rotary [2]

[Total: 25 marks]

Part III Answer one question.

10 The roof truss of a small guard house is triangular in section as shown in **figure 9.** Within the triangle are two supports which form part of the irregular polygon ABCDE. The sides of this polygon are in the ratio AB:BC:CD:DE:EA as 2:3:3:3:3, and the perimeter of the polygon is 12m.

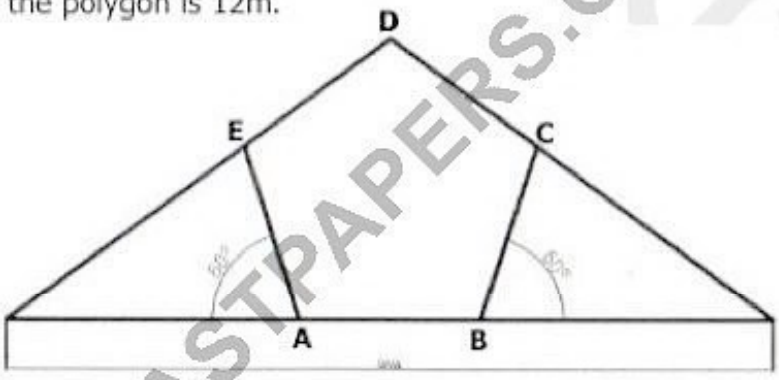


Figure 9

Find the actual span of the roof using the scale of 1:50.

[Total: 30 marks]

ZEDPASTPAPERS.COM

zedpastpapers.com

goto zedpastpapers.com for more

11 In order to help solve the litter problem at Tinenenji Technical Secondary School Tuckshop, learners have decided to design and make litter bins. The litter bins will be attached to the walls of the tuckshop. The chosen solution of the design of the bins is in the shape of a truncated cone as shown in **figure 10.**

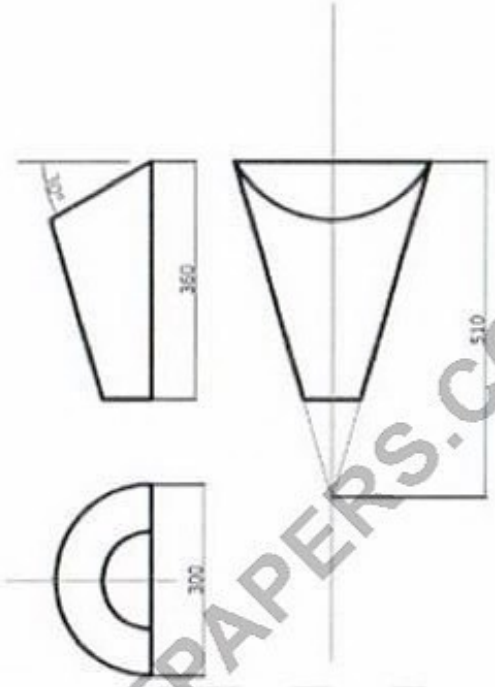


Figure 10

Draw the surface development of the design using a scale of 1:5.

[Total: 30 marks]

ZEDPASTPAPERS.COM
zedpastpapers.com

goto zedpastpapers.com for more